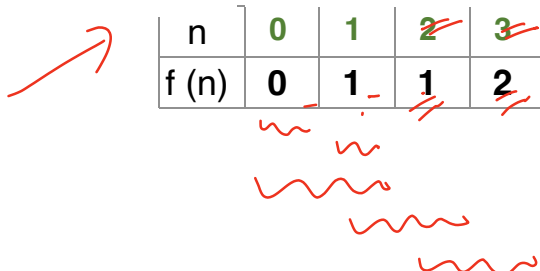


Fibonacci Numbers

Fibonacci Numbers

Given a non-negative number n , design a recursive algorithm to compute the n^{th} Fibonacci number.

i/p: $n=7$
o/p: 13



n	0	1	2	3	4	5	6	7	8	9	...
f(n)	0	1	1	2	3	5	8	13	21	34	...

find n^{th} fib. no.
 $=$

friend? $\Rightarrow$ A $+$ B friend?
 $\Rightarrow$ $n-1^{\text{th}}$ fib. no. $+$ $n-2^{\text{th}}$ fib. no.

it is a fn that finds me n
 $\rightarrow$ $f(n)$ $=$ $f(n-1) + f(n-2)$ $\leftarrow$ rec.
transition A B

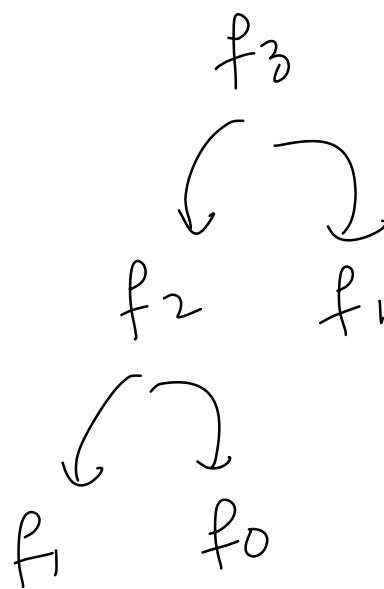
$\left\{ \begin{array}{l} n=0 \\ n=1 \end{array} \right.$

f_4

$f(1) = f(0)$

$f(0) = f(-1)$

) m



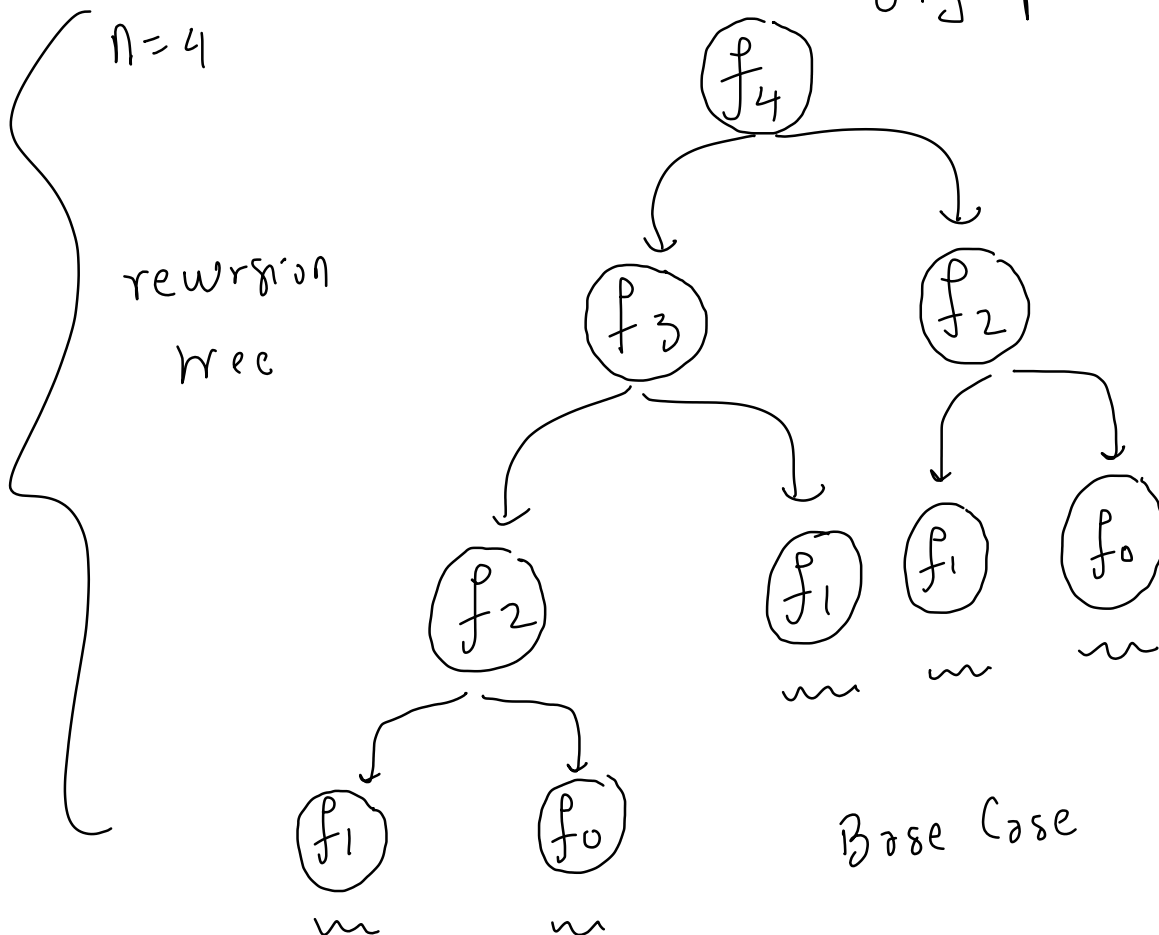
f2

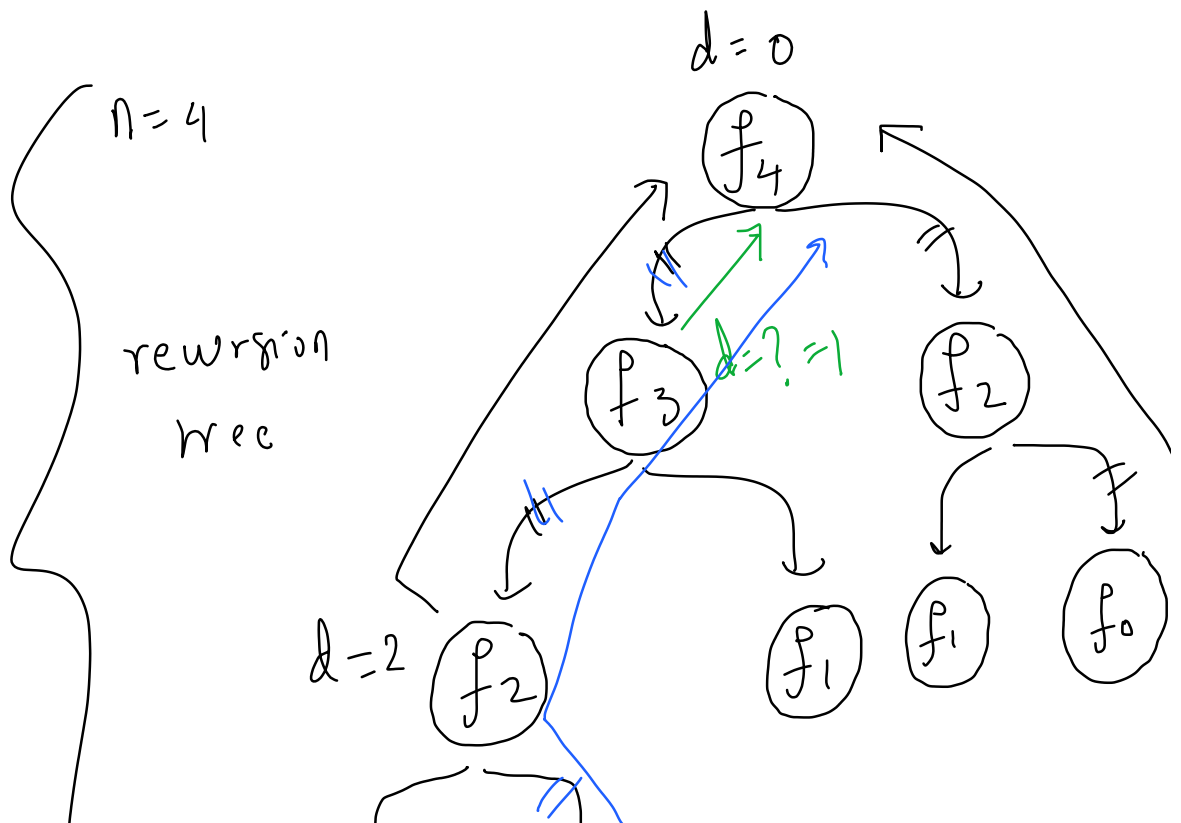
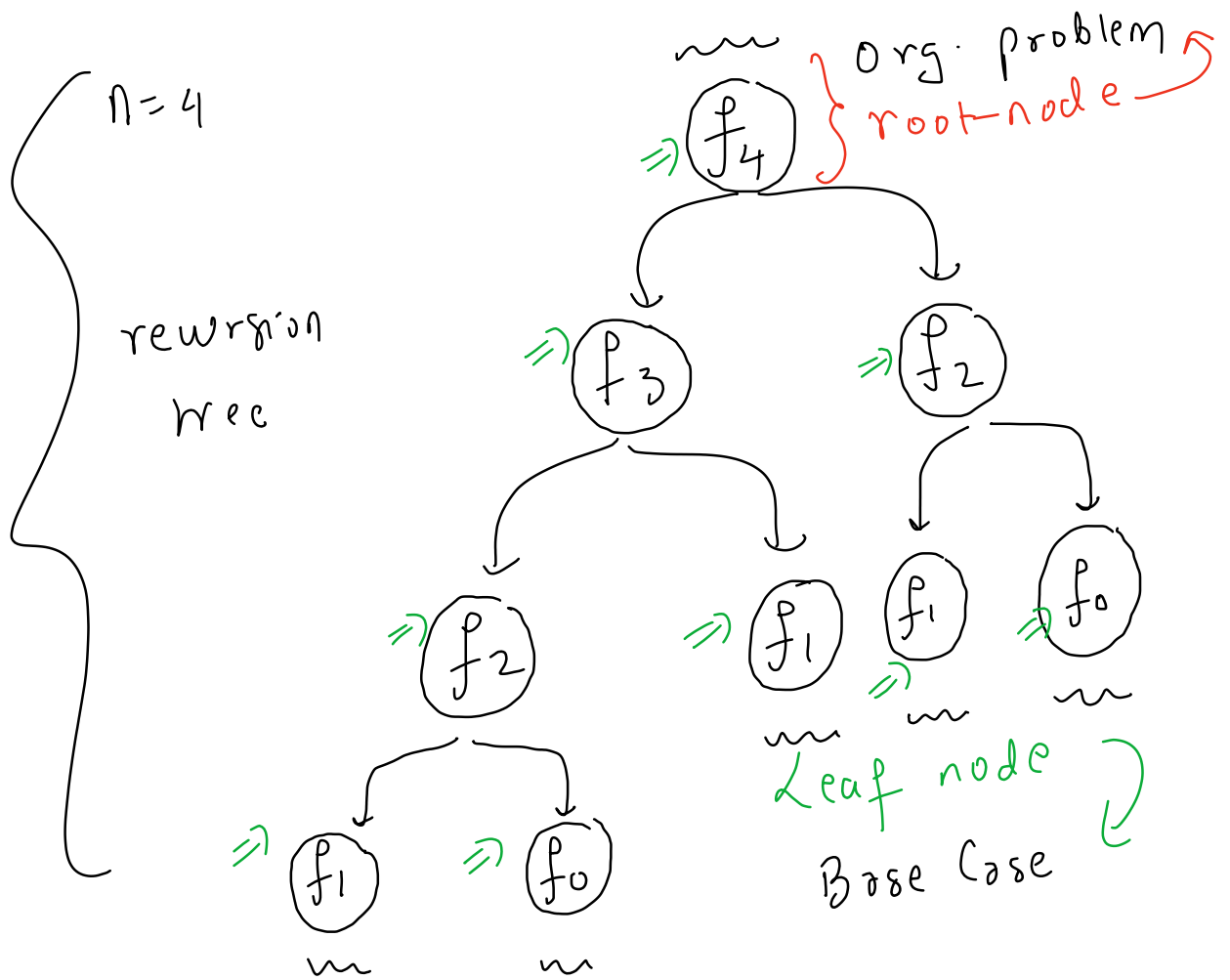
 $f(2) =$

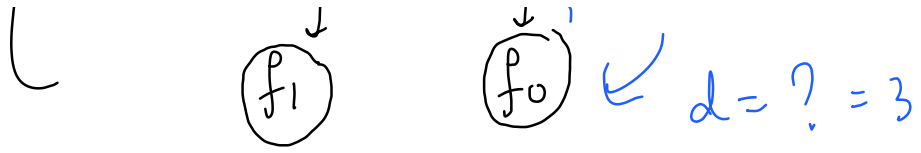
$f(0) = 0$
 $f(1) = 1$

Base Case

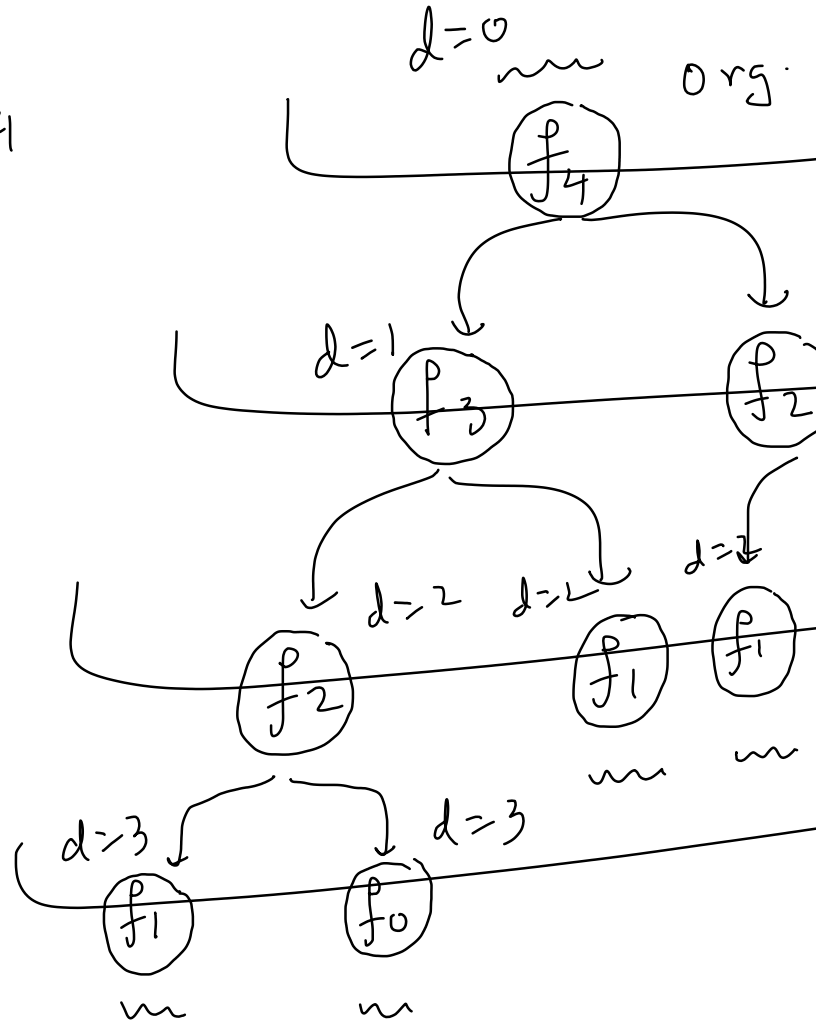
Org. problem







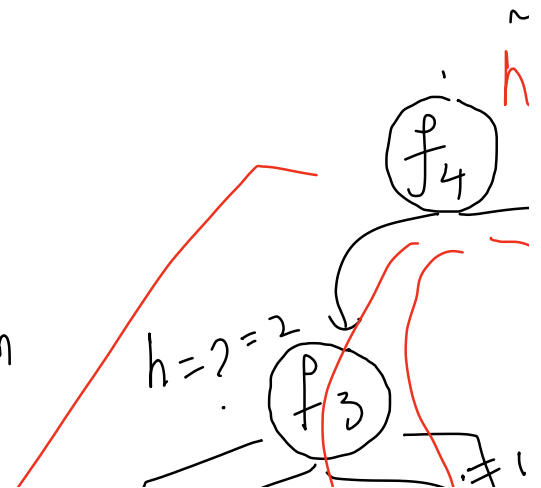
$n = 4$

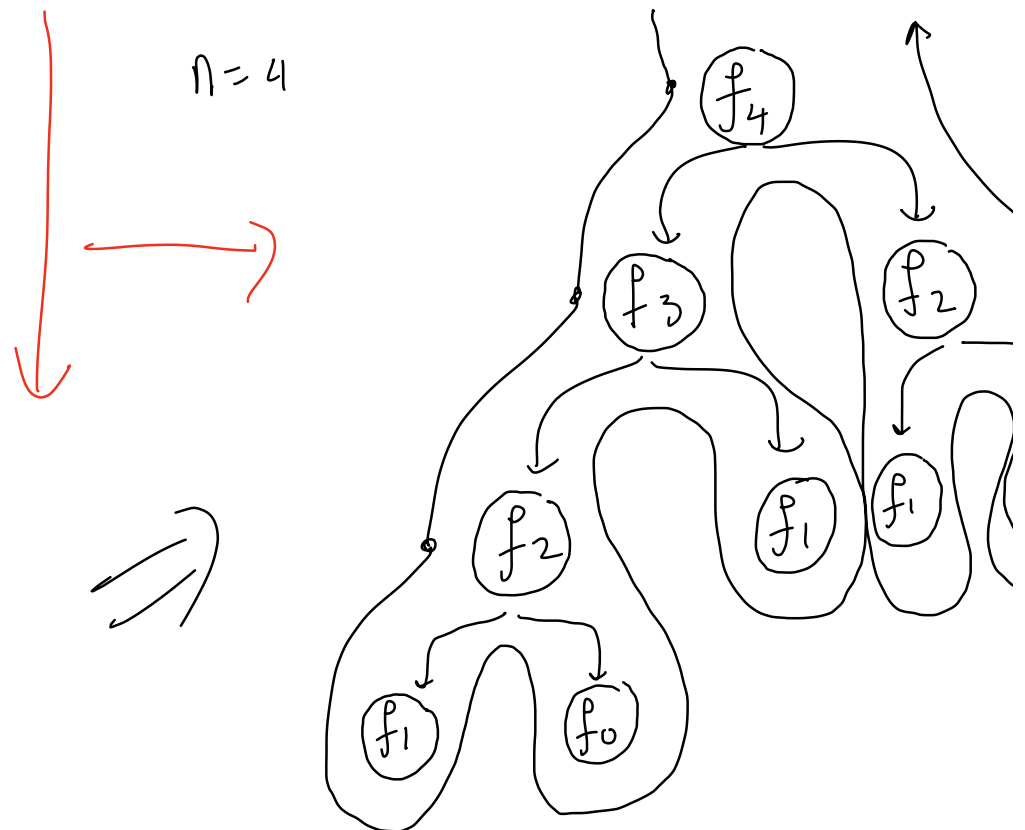
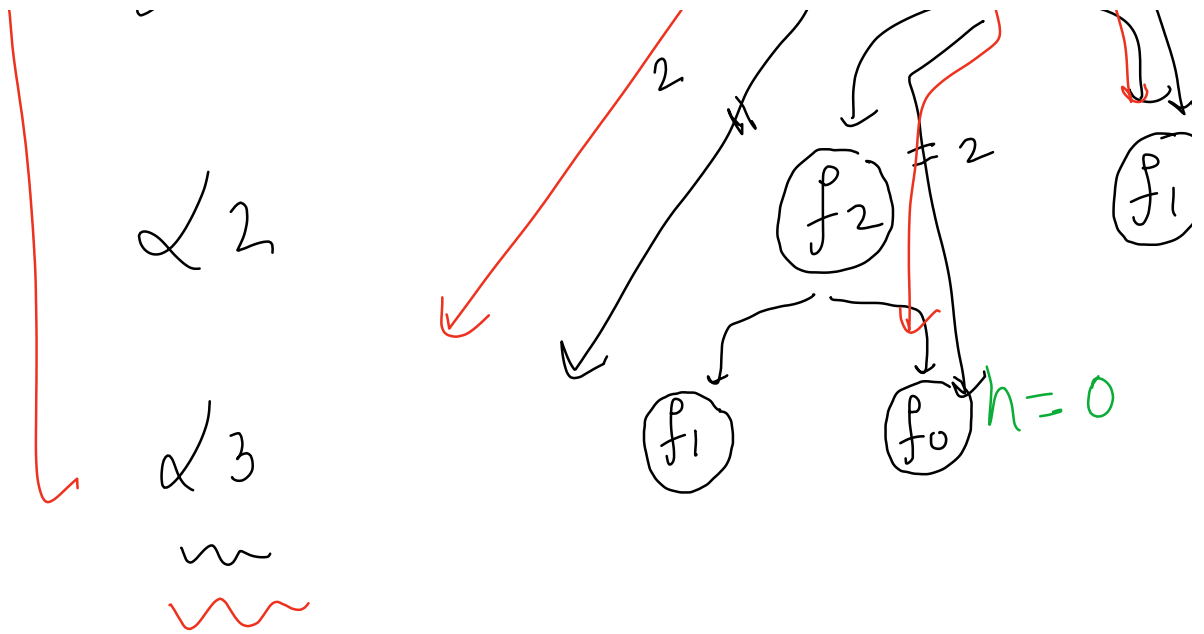


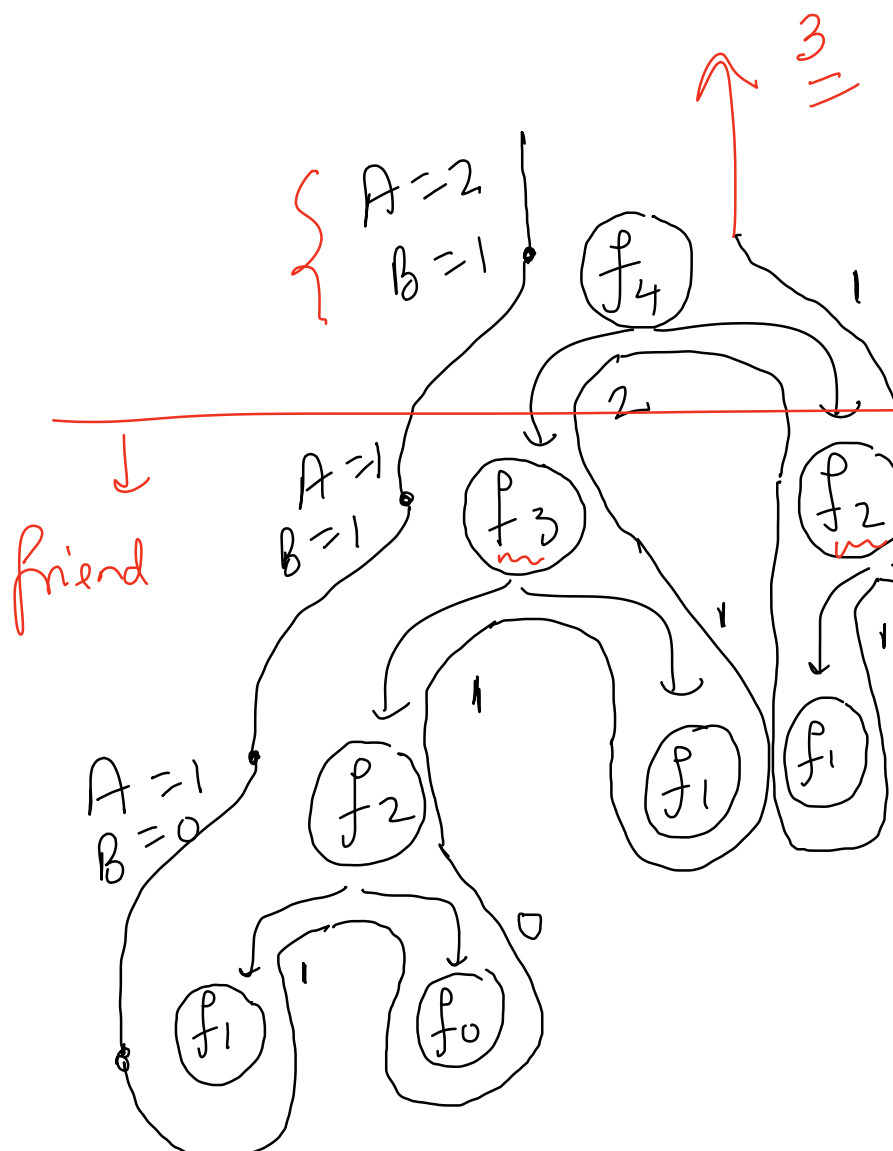
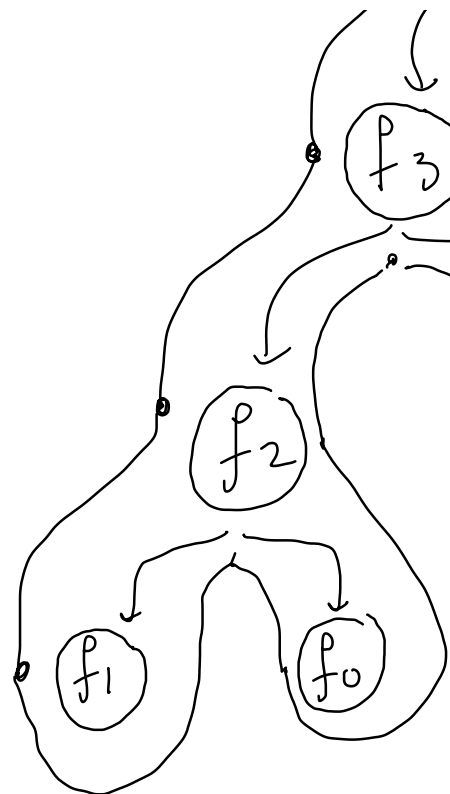
d_0
 d_1

$n = 4$

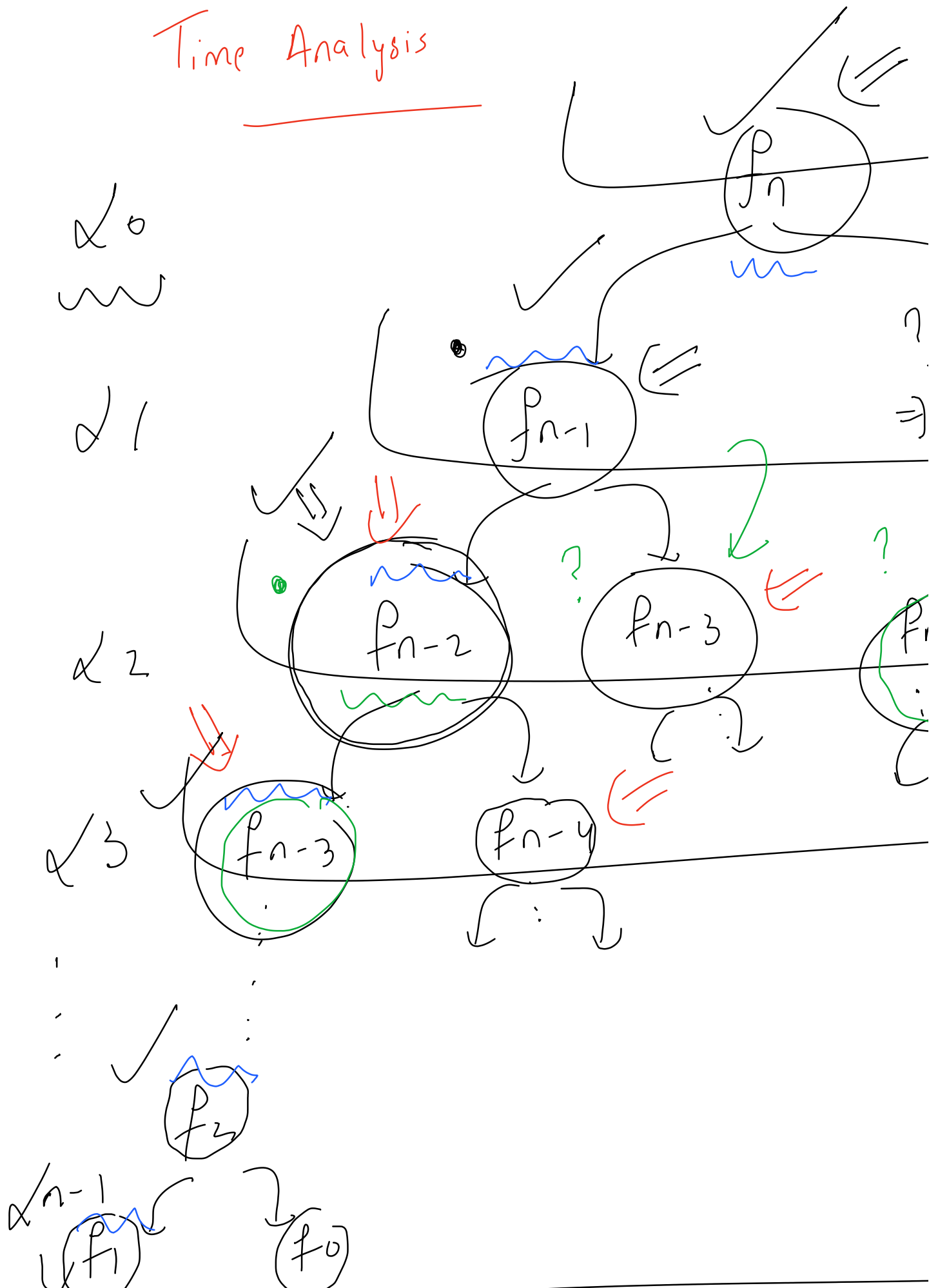
reversion
 nec





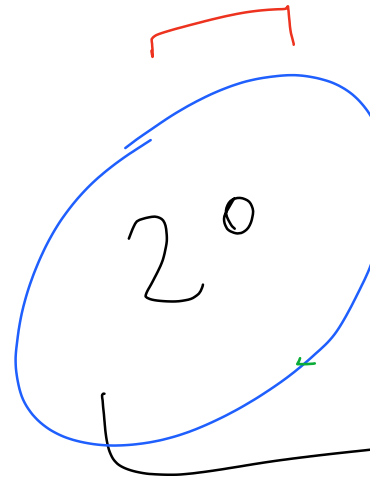


Time Analysis





f_1
⋮
 f_{n-1}
 f_n



$$a = 2^0 = 1$$

$$r = 2$$

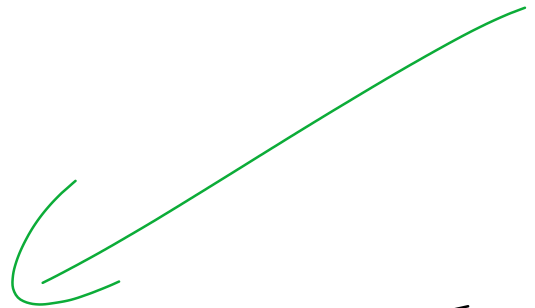
$$\# = n$$



Reversion



optimized
brute force



K



